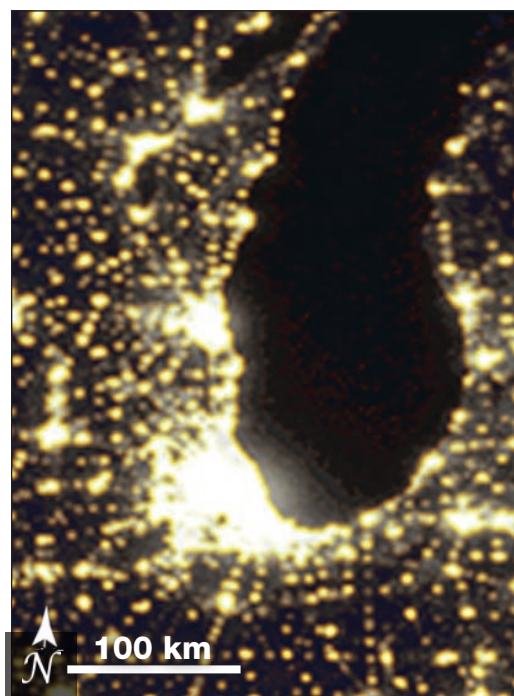




2003 OLS



2012 VIIRS



2016 VIIRS

[Above] *Change Over Time—Chicago, Illinois*. These images show the area surrounding Chicago, Illinois, using nightlight data from the OLS onboard the DMSP released in 2003 (left), the VIIRS DNB in 2012 (middle), and the VIIRS DNB in 2016 (right). When you look at the images side by side, the OLS image appears coarse and blurry compared to the high-precision VIIRS images from 2012 and 2016. The most readily noticeable difference in these nighttime composite views of Chicago and surrounding areas in 2012 and 2016 is lighting along a recently expanded section of Interstate 90. This part of the highway, the Jane Addams Memorial Tollway, links Chicago with Rockford, Illinois, to the northwest.

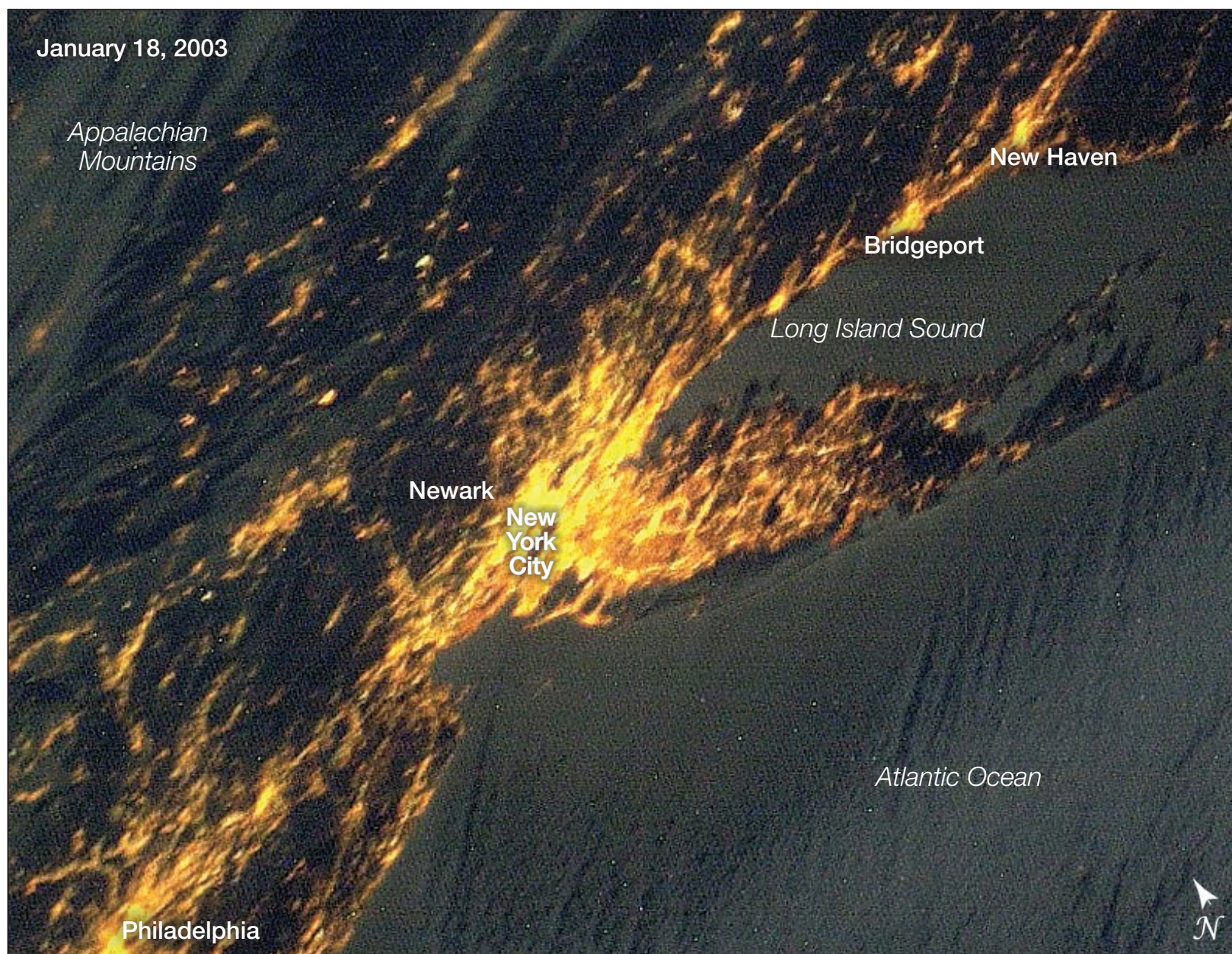
cannot see the surface in the polar regions, although they can observe phenomena in the upper atmosphere. Also, the ISS passes over a given point on Earth every two or three days and at variable times, while, for example, Suomi NPP flies over the same point twice a day at roughly the same time each day. With their complementary capabilities of spatial resolution, timing, and quantitative accuracy, scientists are interested in making joint observations with VIIRS and the ISS.

While scientists use multiple spectral band combinations covering different parts of the electromagnetic spectrum—such as visible and infrared—during nighttime hours, this book focuses mainly on the detection of visible light at night. For comparison, some daytime visible and infrared images will be shown alongside some of the nighttime visible images.

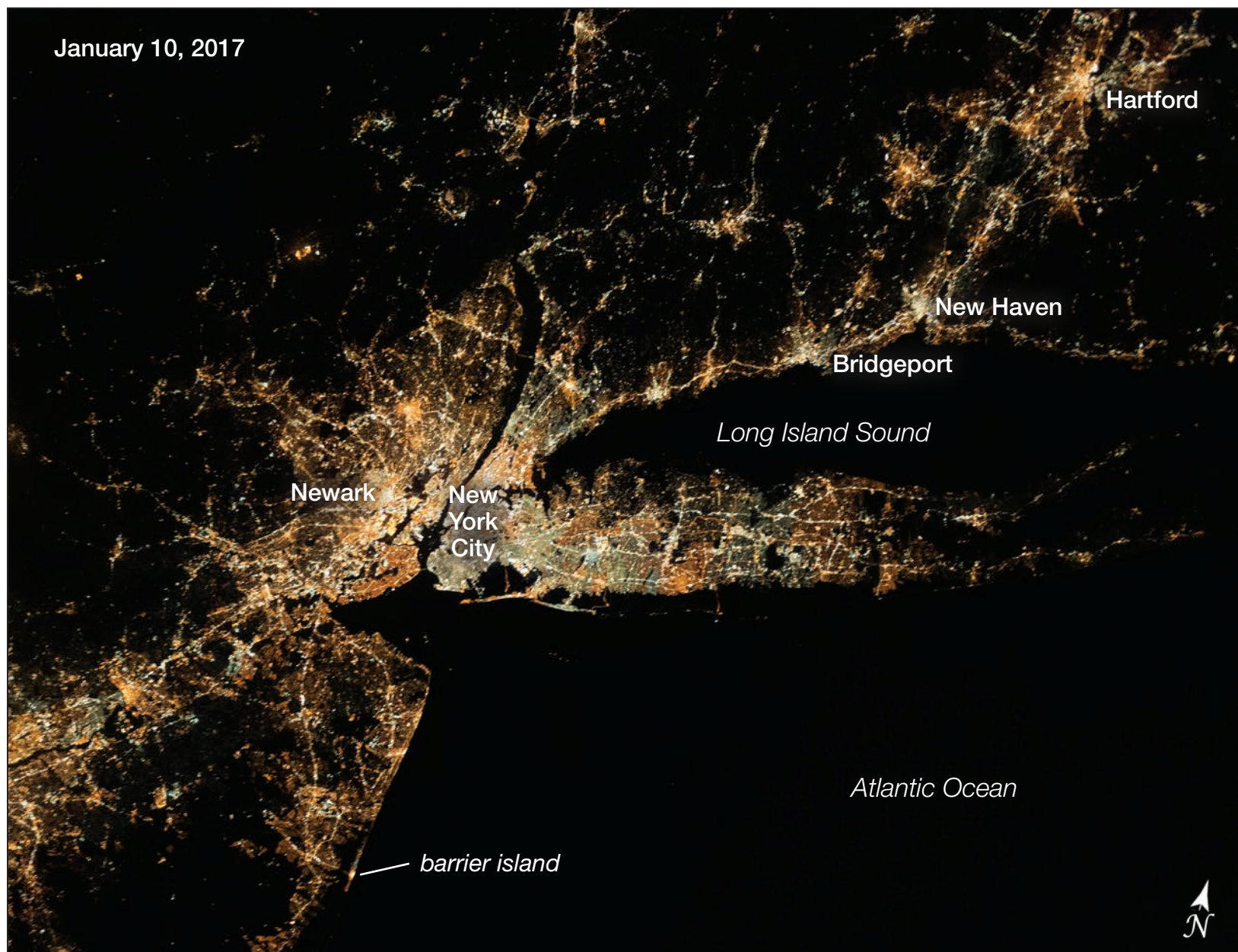


Improvements in ISS Photography—Northeastern United States

This pair of photographs, centered on New York City in the northeastern United States, shows improvements in camera technology used on the ISS over a 14-year time period. The image on the left was taken on January 18, 2003, with a Nikon D1 digital camera (3 megapixels), while the image on the right was taken on January 10, 2017, with a Nikon D4 digital camera (16 megapixels).



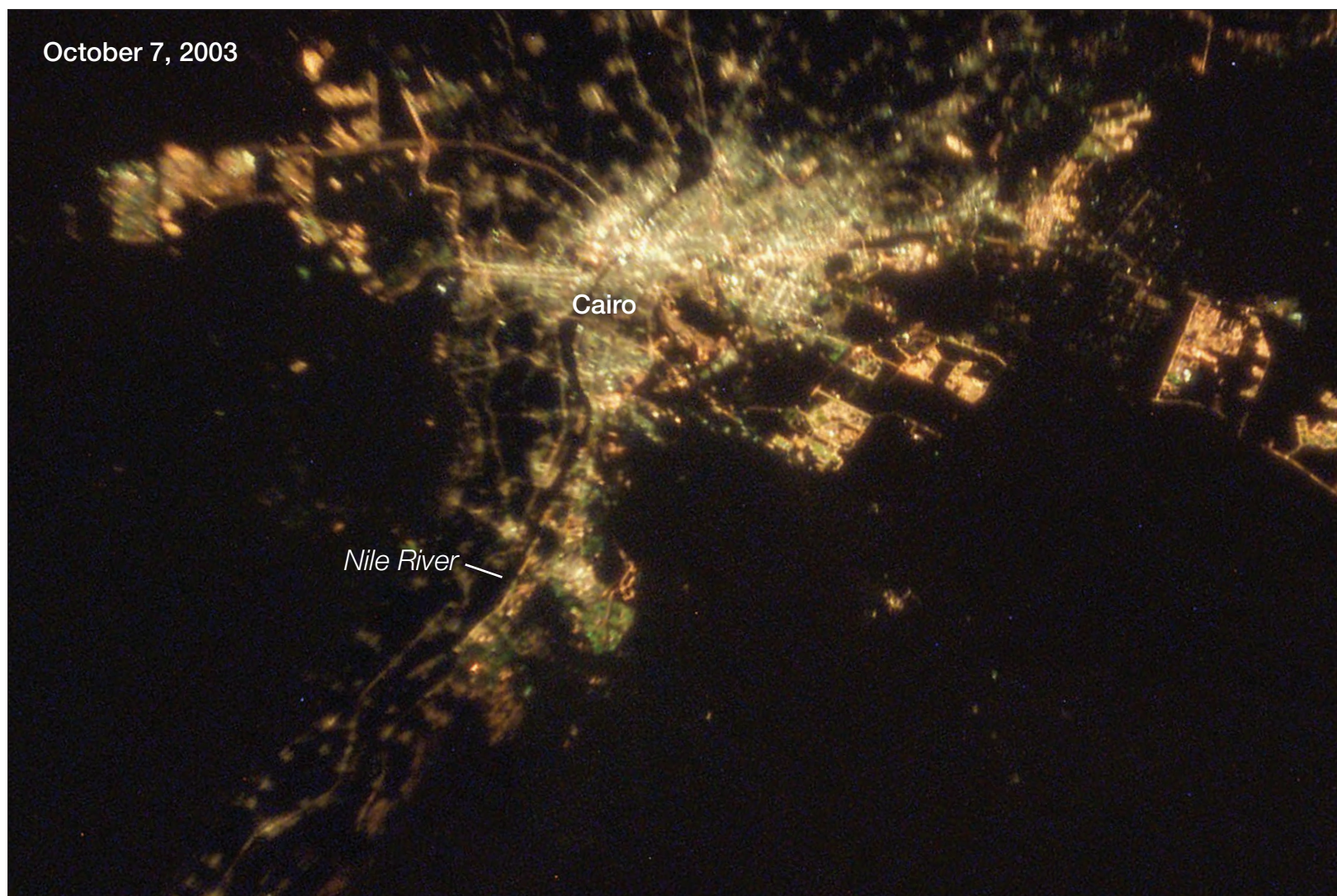
The 2003 image was taken from a vantage point well to the northeast of the cities, with the camera pointed westward back towards New York City and the coast. The result is that the perspective is distorted but still recognizable. Low clouds have formed over the waters of the Atlantic and have settled into some of the valleys of the Appalachian Mountains to the northwest.





Coming into Focus—Cairo, Egypt

This pair of photographs, centered on Cairo, Egypt, shows improvements in camera technology used on the ISS between October 7, 2003 (left) and December 10, 2014 (right). The 2003 photo was taken with a Kodak DCS 760c electronic still camera (6.3 megapixels), while the 2014 image was taken with a Nikon D4 digital camera (16 megapixels).





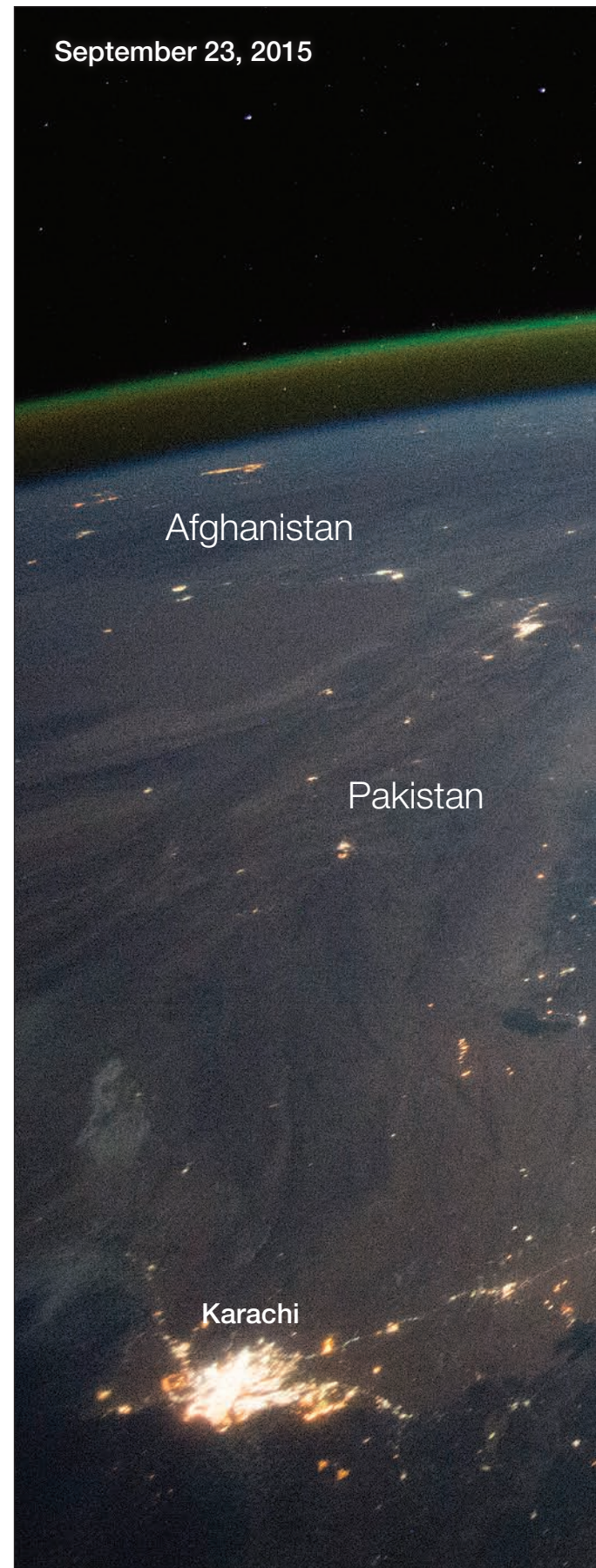
Where Do Nightlights Come From?

The night side of Earth twinkles with light, easily observed by orbiting satellites. The most obvious source comes from city lights, but as one looks closer, much more becomes discernible. Even at some distance from human settlements, light still shines: Wildfires and volcanoes rage; gas flares glow like candles; auroras dance across the polar skies; moonlight and starlight reflect off surface water, snow, and deserts; and clouds are easily seen. Even the air and ocean sometimes glow.

The “Nature’s Light Shows” and “Human Light Sources” sections of this book feature images of natural and anthropogenic light sources to illustrate how scientists use nightlight data to study our changing planet and how decision makers, in turn, can use the knowledge gained for public benefit. Some of these applications include forecasting a city’s energy use and carbon emissions, eradicating energy poverty and fostering sustainable energy development, providing immediate information when disasters strike, and monitoring the effects of conflict and population displacement.

[Right] An astronaut onboard the International Space Station took this nighttime panorama while looking north across Pakistan’s Indus River valley. The port city of Karachi is the bright cluster of lights facing the Arabian Sea, which appears completely black. City lights and the dark color of dense agriculture closely track with the great curves of the Indus valley. For scale, the distance from Karachi to the foothills of the Himalaya Mountains is 720 miles (1,160 kilometers). This photograph shows one of the few places on Earth where an international boundary can be seen at night. The winding border between Pakistan and India is lit by security lights that have a distinct orange tone.

September 23, 2015





Beyond City Lights—Java Sea

The visible lights produced by modern civilization are often obscured by the more intense light of the Sun during the day. However, as this photograph—taken over the Java Sea by a member of the Expedition 42 crew onboard the ISS—reveals, at night these “nighttime lights” become visible. To the discerning eye, these lights reveal intricate details of both natural phenomena and the unmistakable footprints of human civilization. The photo, taken on October 20, 2014, captured light from roads and cities, fishing boats, lightning, airglow, and even a few gas flares. North of the city of Surabaya, the capital of East Java province, large numbers of boat lights dot the darkened Java Sea. Many of these vessels are likely fishing boats using bright lights to attract squid and other sea life. According to one estimate based on satellite observations, there were over 500 fishing boats in the area on a single night in September 2014. The same study pointed out that two of the brighter points of light in the area are gas flares emanating from offshore oil infrastructure. In addition to the human activity, the photograph also showcases a few natural splashes of light and color. The blue-white patch on the far left is a flash of lightning, a common sight for astronauts. The glowing line along Earth’s limb—or edge of the atmosphere—is airglow, a type of light that is produced by chemical reactions in the upper atmosphere. While airglow can also be green or blue, red airglow like this comes from excited oxygen atoms at heights of about 90 to 190 miles (150 to 300 kilometers).

October 20, 2014







Moonlight, Old Lights, and New Lights—Gulf of Mexico

This wide-angle, nighttime photograph was taken on February 11, 2015, by ISS astronauts looking southeastward over the Gulf of Mexico. Moonlight reflects diffusely off the waters of the Gulf, making the largest illuminated area in the image. The sharp edges of light patterns from coastal cities trace the long curve of the shoreline from

February 11, 2015

